CSE558 · DATA SCIENCE — PROBABILITY & INFERENCE

Random Variables & the Case for Testing

TODAY'S ARC

Recap → Distributions → Coupon Collector → CLT → The Mystery

Supratim · Monsoon 2026

Claude & Gemini helped with slides.

RECAP

Where we left off

1

Describing shape

Variance, standard deviation, mean absolute deviation, skewness, kurtosis.

2

Cleaning data

Outliers, missing values, noise. How to reason with the dataset before you dig deeper.

3

Today's question

None of that works for data you haven't collected yet. How do you reason about a population from just a sample?

A different kind of question

WHAT YOU COULD ALREADY DO

“Here are 12 exam scores. What's their mean? Are they skewed? Any outliers?”

The data is sitting right there. You just describe it.

WHAT YOU'RE ABOUT TO LEARN

“A friend claims she can guess a card's suit better than chance. She got 52/200 right. Is that real, or luck?”

You'll never see every guess she could ever make. You have to reason from a sample to a truth you can't fully observe.

PREREQUISITE REFRESHER

Probability space

THREE PARTS

Sample space Ω

Every possible outcome, all together.

Events

Any subset of Ω you care about.

Probability function P

Assigns a likelihood to each event.

The rules: $P(\Omega) = 1$, $P(\emptyset) = 0$, and $0 \le P(A) \le 1$ for every event A.

EXAMPLE — ROLLING A DIE

$\Omega = \{1, 2, 3, 4, 5, 6\}$

Event "even" $= \{2, 4, 6\}$

$P(\text{even}) = 3/6 = 0.5$

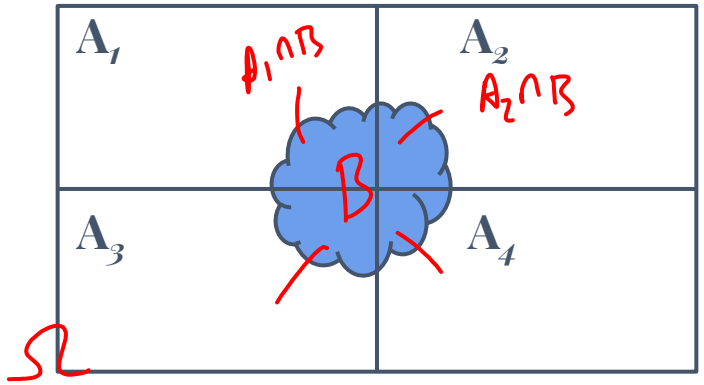
Event Space/σ-algebra

Let $\Omega \neq \emptyset$ be a set. Let $\boldsymbol{P}(\Omega)$ be its power set.

Then $\boldsymbol{F} \subseteq \boldsymbol{P}(\Omega)$ is called a σ-algebra, if it satisfies:

1. $\boldsymbol{F}$ contains sample space, i.e., $\Omega \in \boldsymbol{F}$
2. $\boldsymbol{F}$ is closed under complement:
    a. If $A \in \boldsymbol{F}$ then $A^c := \{\Omega \setminus A\} \in \boldsymbol{F}$
3. $\boldsymbol{F}$ is closed under countable unions:
    a. If $A_i \in \boldsymbol{F}$ for $i = 1, 2, \ldots$ then $\bigcup_i A_i \in \boldsymbol{F}$

Law of Total Probability



$$\mathbf{P}(\mathbf{B}) = \sum_{i=1}^{n} \mathbf{P}(\mathbf{B} \cap \mathbf{A}_i) = \sum_{i=1}^{n} \mathbf{P}(\mathbf{B}|\mathbf{A}_i) \cdot \mathbf{P}(\mathbf{A}_i)$$

for any partition {A_1, A_2, …, A_n} of the sample space.

EXAMPLE

3 machines make 50% / 30% / 20% of output, with defect rates 2% / 3% / 5%.

What is the probability of defect?

Intuition

Break a hard-to-compute probability into easier conditional pieces, then weight each piece by how likely it is to happen in the first place.

Random variables

A function that maps every outcome in the sample space to a number.

Why bother??

$X \begin{cases} S_1 \to 1 \\ S_2 \to 2 \\ \vdots \\ S_n \to n \end{cases}$

$E[X] = \sum_{i=1}^{n} i \cdot P_X(X = i)$

$Var(X) = E[(X - E(X))^2]$

EXAMPLE: COUNTING NUMBER OF HEADS IN A TOSS OF 3 COINS

How many elements in the sample space?

HHH	$\to X = 3$
HHT, HTH, THH	$\to X = 2$
HTT, THT, TTH	$\to X = 1$
TTT	$\to X = 0$

Linearity of expectation

E[aX] = a·E[X] and E[X + Y] = E[X] + E[Y]

$$\mathbf{E}[\mathbf{X}+\mathbf{Y}] = \sum_i \sum_j (i+j)\cdot \mathbf{P}(\mathbf{X}==i \cap \mathbf{Y}==j)$$

$$= \sum_i i \sum_j \mathbf{P}(\mathbf{X}==i \cap \mathbf{Y}==j) + \sum_j j \sum_i \mathbf{P}(\mathbf{X}==i \cap \mathbf{Y}==j)$$

$$= \sum_i i \cdot \mathbf{P}(\mathbf{X}==i) + \sum_j j \cdot \mathbf{P}(\mathbf{Y}==j)$$

QUICK CHECK — VOTE TRUE OR FALSE

Does linearity hold for variance too?

$Var(X+Y) = E[((X+Y) - E(X+Y))^2]$

$=$

$E(XY)$

$E(X)\ E(Y)$

QUICK CHECK

Does linearity hold for variance too?

Unlike expectation, variance does NOT add freely. In general:

$$\mathbf{Var[X+Y] = Var[X] + Var[Y] + 2 \cdot E[(X-E[X]) \cdot (Y-E[Y])]}$$

When X and Y are independent, then?



DISTRIBUTIONS

Bernoulli random variable

For SINGLE trial with exactly two outcomes: success (1) or failure (0), with success probability p.

$$P(X = 1) = p \qquad P(X = 0) = 1 - p$$

EXPECTATION & VARIANCE

$$E[X] = p \qquad Var[X] = p(1-p)$$

Since X only takes values 0 and 1, E[X] is literally just the probability of getting a 1.

The Binomial random variable

Counts the number of successes in n independent trials, each with success probability p. Denoted B(n, p).

$$P(X = i) = C(n,i) \cdot p^{i} \cdot (1-p)^{n-i}$$

where i = 0, 1, …, n

EXPECTATION

$$E[X] = np$$

By linearity of expectation — sum n Bernoulli means, each worth p.

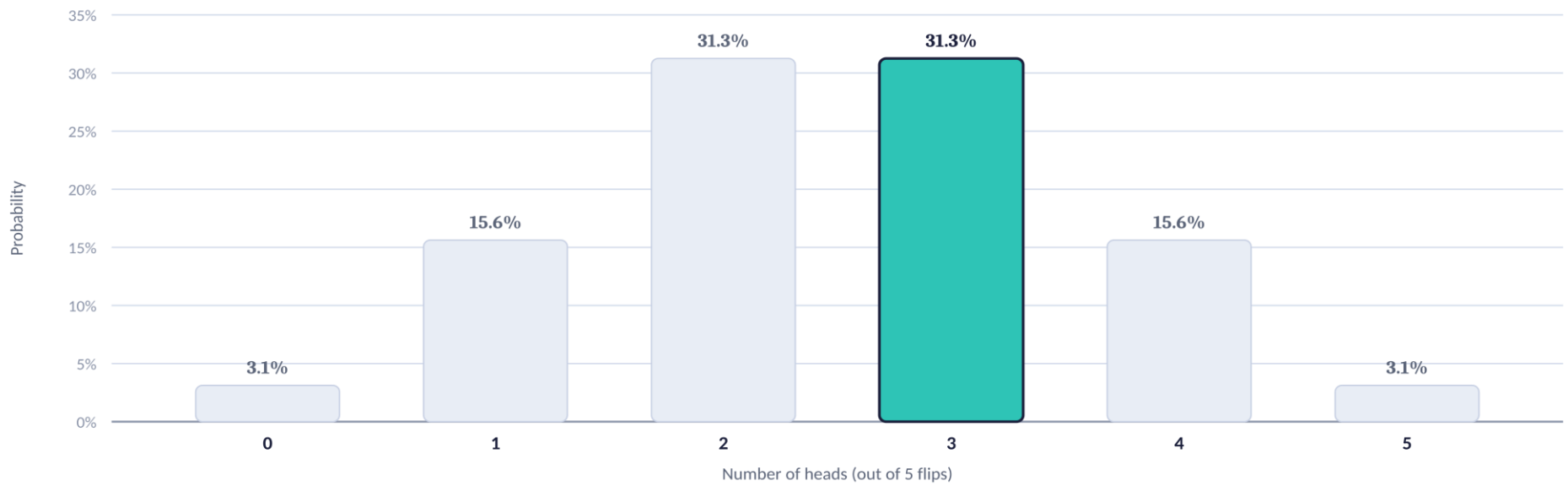
It's an aggregation of Bernoulli trials

Each individual trial is a Bernoulli(p) — a single coin flip. Sum n of them together, and you get a Binomial(n, p).

TRY

Compute a binomial probability

Flip a fair coin 5 times. What's the probability of exactly 3 heads? n=5, p=0.5, i=3.



$C(5,3) \cdot 0.5^3 \cdot 0.5^2 = 10 \times 0.125 \times 0.25 = 0.3125$ — matches the highlighted bar.

The Geometric random variable

$E[X] = 1p + 2p(1-p) + 3p(1-p)^2 + \dots$

Counts the number of trials until the FIRST success, each trial independent with success probability p.

$$\mathbf{P(X = i) = (1-p)^{i-1} \cdot p}$$

where i = 1, 2, 3, … (unbounded)

INTUITION

$(1-p)^{i-1}$ is the probability of failing i−1 times before, succeeding with probability p.

$$\mathbf{E}[X] = 1p + 2(1-p)p + 3(1-p)^2p + 4(1-p)^3p + \dots$$

$$-(1-p)\mathbf{E}[X] = 1(1-p)p + 2(1-p)^2p + 3(1-p)^3p + \dots$$

$$p\mathbf{E}[X] = p + (1-p)p + (1-p)^2p + \dots$$

$$\mathbf{E}[X] = 1 + (1-p) + (1-p)^2 + \dots = \frac{1}{1-(1-p)} = \frac{1}{p}$$

TRY IT

Waiting for a six

THE SETUP

You roll a fair six-sided die repeatedly, waiting for your first 2. So p = 1/6.

QUESTION 1 — expected wait

On average, how many rolls until your first 2? $\frac{1}{p} = 6$

QUESTION 2 — a specific probability

What's the probability your first 2 lands on exactly the 3rd roll? $\left(\frac{5}{6}\right)^2 \frac{1}{6}$

DON'T CONFUSE THEM

Binomial vs. Geometric

BINOMIAL — fixed n trials, count every success



All 10 trials happen. Count = 4 successes.

GEOMETRIC — wait for the very first success



Stop counting the moment you succeed. Wait = 5 trials.

Binomial

"Flip a coin 10 times — how many heads?"

Bounded: 0 to n successes.

Geometric

"Flip until it's heads — how many flips?"

Unbounded: 1 to ∞.

QUICK CHECK

Which distribution fits?

Number of defective items in a batch of 50	Binomial
Coin flips until the first tails	Geometric
Throws on target out of 10 attempts	Binomial
Customers browsed until the first one buys	Geometric
Correct guesses on a 20-question quiz	Binomial

APPLYING IT

The Coupon Collector Problem

A cereal box has one of k different toy prizes inside, chosen uniformly at random each time. If you keep buying boxes, how many do you need on average to collect all k toys?

The key insight

BINOMIAL OR GEOMETRIC?

Getting your very first toy is easy!

But once you have most of the set, most boxes just give you a duplicate.

Each new toy gets harder to find than the last.

APPLYING IT

Breaking it into geometric pieces

Let X_i = boxes needed to find the i-th NEW toy, after you already have i−1 types. With k=4 toys total:

Toy #	P(next box is new)	Expected wait $E[X_i]$
1st toy	4/4 (any box is new)	$\mathbf{E[X_1] = 4/4 = 1}$
2nd toy	3/4	$\mathbf{E[X_2] = 4/3 \approx 1.33}$
3rd toy	2/4	$\mathbf{E[X_3] = 4/2 = 2}$
4th toy	1/4 (only 1 left to find)	$\mathbf{E[X_4] = 4/1 = 4}$

Each row's probability shrinks as fewer new toys remain — which is exactly why the last toy is the hardest to find.

APPLYING IT

Analysis

ADD UP ALL FOUR WAITS

E[Total] = 1 + 4/3 + 2 + 4 = 8.33 boxes

On average, you need about 8–9 boxes to collect all 4 toys, roughly double the number of types.

$$\mathbf{E}\left[\sum_{i=1}^{k} \mathbf{X}_i\right] = \sum_{i=1}^{k} E(X_i) = 1 + \frac{1}{\frac{k-1}{k}} + \frac{1}{\frac{k-2}{k}} + \cdots + \frac{1}{\frac{1}{k}}$$

$$= 1 + \frac{k}{k-1} + \frac{k}{k-2} + \cdots + k = 1 + k\sum_{i=1}^{k-1} \frac{1}{k-1} = 1 + k\log k$$

$$\mathbf{E}\left[\sum_{i=1}^{k} \mathbf{X}_i\right] = \sum_{i=1}^{k} \mathbf{E}[\mathbf{X}_i] = \sum_{i=1}^{k} \frac{1}{p_i} = \sum_{i=1}^{k} \frac{k}{k-i+1} = k\log(k)$$

The Poisson random variable

Counts how many rare, independent events happen in a fixed interval of time or space, at a constant average rate λ.

$$P(X=k) = \frac{(\lambda^k e^{-\lambda})}{k!}$$

where k = 0, 1, 2, … (unbounded)

Where you'll see it

Customers arriving per hour, typos per page, earthquakes per year, emails per day — rare events scattered across a fixed window.

EXPECTATION & VARIANCE

$$E[X] = \lambda$$

$$Var[X] = \lambda$$

The mean and the variance are the exact same number.

The Exponential random variable

Measures the waiting TIME until the next event in a Poisson process — the continuous cousin of Geometric.

$$P(X=k) = (\lambda e^{-\lambda k})$$

for $x \geq 0$ — a probability DENSITY, since time is continuous

Memoryless

Like Geometric, however long you've already waited tells you nothing about how much longer you'll wait.

EXPECTATION & VARIANCE

$$E[X] = 1/\lambda$$

The same 1/p as Geometric — time just replaced trial count.

SAME PROCESS, TWO QUESTIONS

Poisson vs. Exponential

Poisson

DISCRETE count of events in a fixed window. “How many calls in the next minute?” Answer: a whole number, 0 to ∞.

Exponential

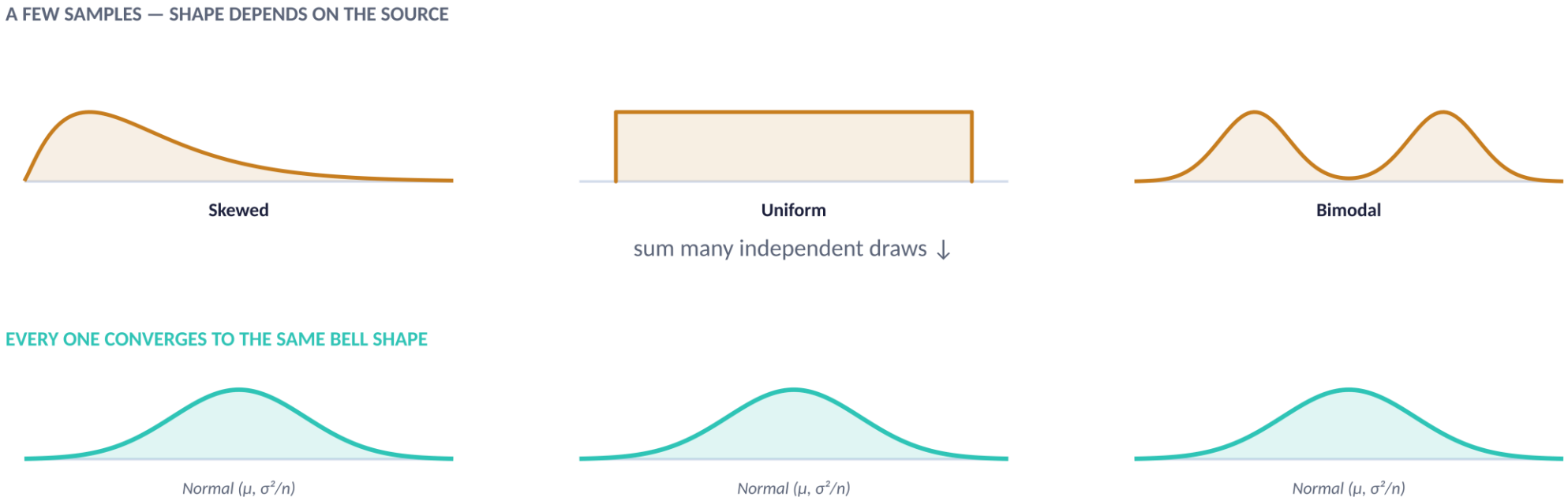
CONTINUOUS time until the next event. “How long until the next call?” Answer: any non-negative real number.

SAME λ, SAME UNDERLYING PROCESS

Poisson is to Binomial as Exponential is to Geometric — the same discrete-count vs. continuous-wait pairing, just with rare events over an interval instead of a fixed number of trials.

The Central Limit Theorem

Sum (or average) many independent samples, and the result tends toward a normal distribution — no matter what the original distribution looked like.



DISCUSSION

Why the CLT is the backbone of everything ahead

WHERE YOU'VE ALREADY SEEN IT

- Exam scores across a large class tend to look bell-shaped
- Polling results across many independent respondents
- Average height, weight, or reaction time in a population

WHY IT MATTERS

Many hypothesis test ahead assumes a sample mean behaves normally.

THE MYSTERY

Your friend's suspicious claim

A friend claims she can guess the suit of a randomly picked card correctly more than 25% of the time — better than pure chance (which would be exactly 1/4).

THE FIRST TEST

You pick 200 random cards. She correctly guesses more than 52 of them (> 26%).

DO YOU BELIEVE HER?

52 out of 200 is more than chance would predict. But is it enough or she was lucky?



THE MYSTERY

Does more evidence change your mind?

THE SECOND TEST

You pick 100 more cards. She correctly guesses more than 27 of them (> 27%).

Now what?

WHAT WOULD ACTUALLY CONVINCE YOU?

A bigger sample? A higher percentage? Consistency across both tests?

To be continued: next lecture gives you the exact tool to answer this rigorously.



STUDY MATERIAL

Reference

Probability & Computing: Chapter 2 by Mitzenmacher & Upfal

Slides